1. Executive Summary

The assessed Terraform on `main` branch across three local repositories shows the active AWS footprint is concentrated in EKS-based platform infrastructure, ingress/WAF, messaging (SNS/SQS), encryption/secrets, EFS, and backup storage, with the workload service code repo containing no Terraform under the requested paths. For a latency-sensitive API workload with 99.9% availability, RTO 4h, RPO 30m, and SOC2 plus regional residency constraints, the directional economics and migration risk profile favour Azure as the primary target for lower modelled 30-day run-rate and lower platform porting friction, with GCP as a viable alternative where GKE and Pub/Sub strengths are preferred. Recommended path: phased Azure-first migration with staged DR rehearsal and policy-as-code parity gates before production cutover.

2. Source Repository Inventory

Source type	Repository path	Branch used	Terraform scope scanned	File count (.tf/.tfvars)	Notes
local-path	/Users/srimanta.singh/IdeaProjects/hxp-transform-service	main	src/, infra/, terraform/	0	No Terraform files found in requested directories
local-path	/Users/srimanta.singh/IdeaProjects/terraform-aws-hxts-environment	main	src/, infra/, terraform/	21	Queue/KMS/topic integration and per-environment tfvars
local-path	/Users/srimanta.singh/IdeaProjects/tf-cfg-hxts-infrastructure	main	src/, infra/, terraform/	67	Core EKS/shared services/network/security/backup stack

3. Source AWS Footprint

Resource group	Key AWS services found	Notes
Compute	EKS managed node groups/addons, Karpenter	`eks_instance_type = m5.xlarge` in all environment tfvars; desired_size 3 nodes
Networking	ALB/NLB, Route53, VPC endpoint service, Security Groups	Public ingress + private NLB target groups; PrivateLink endpoint service exposed
Data	EFS (`throughput_mode = elastic`, KMS-backed), S3 (Velero backup + cross-region replication)	Multi-region EFS and S3 replication; storage class `STANDARD`
Messaging	SQS (14+ named queues), SNS (2 topics: request/reply), subscriptions, queue policies	Fan-out + per-engine queues with KEDA scalers
Identity/Security	IAM roles/policies, KMS (multi-region), Secrets Manager, WAFv2	KMS `multi_region=true`; WAF log retention 7 days; secrets versioned
Observability	CloudWatch log groups (EKS, WAF), Datadog monitors/dashboards	CW retention 14 days; Datadog container-ready monitors per region
Storage/Backup	Velero S3 bucket + cross-region replica bucket	Backup region variable present (e.g. `eu-west-1` mirrors `eu-central-1`)

4. Service Mapping Matrix

AWS service	IaC-provisioned tier/family	Azure equivalent (matched tier)	GCP equivalent (matched tier)	Porting notes
Amazon EKS managed nodes	`m5.xlarge` (4 vCPU / 16 GB)	AKS node pool `Standard_D4s_v5`	GKE node pool `n2-standard-4`	Rework IRSA → Workload Identity; Karpenter → Karpenter/CAST AI/Cluster

AWS service	IaC-provisioned tier/family	Azure equivalent (matched tier)	GCP equivalent (matched tier)	Porting notes
				Autoscaler equivalent
EKS control plane	Managed (tier not specified in IaC)	AKS control plane (included in AKS)	GKE Standard control plane	Cost model differs; AKS control plane free at Standard tier
Application Load Balancer	Tier not specified in IaC	Azure Application Gateway v2	Cloud Load Balancing (ext. HTTP(S))	Listener/rule migration; health probe and certificate management
AWS WAFv2	Tier not specified in IaC	Azure WAF (App Gateway policy)	Cloud Armor	Managed rule set translation required; monitor mode burn-in
Amazon SQS	Tier not specified in IaC (Standard queue inferred)	Azure Service Bus Standard queues	Pub/Sub standard	14 distinct queue names; DLQ and visibility timeout semantics
Amazon SNS	Tier not specified in IaC	Service Bus Topics / Event Grid	Pub/Sub topics	Filter policy equivalence; cross-account publish must be re-modelled
AWS KMS	Symmetric CMK, `multi_region = true`	Azure Key Vault Keys (Premium/HSM if required)	Cloud KMS CMEK	Key rotation, replication policy, and access policy model differ
AWS Secrets Manager	Versioned secrets	Azure Key Vault Secrets	Secret Manager	Rotation lambda → managed rotation; app identity bindings must change
Amazon EFS	`throughput_mode = elastic`, KMS-backed	Azure Files Premium (NFS 4.1)	Filestore Enterprise	POSIX parity OK; elastic throughput → provisioned IOPS sizing required
Amazon S3 (Velero)	`STANDARD` storage class, cross-region replication	Azure Blob Hot + GRS/RA-GRS	GCS Standard + dual-region	Velero plugin swap; IAM and bucket policy re-implementation
Route53 records	Tier not specified in IaC	Azure DNS	Cloud DNS	TTL, failover, and health check parity checks
VPC Endpoint Service (PrivateLink)	Tier not specified in IaC	Azure Private Link service	Private Service Connect	Endpoint consumer re-approval model; NLB front-end required on Azure
Karpenter node provisioner	Tier not specified in IaC	AKS Karpenter (preview) / KEDA	GKE Autopilot / Cluster Autoscaler	Native availability varies by cloud; tuning required

5. Regional Cost Analysis (Directional)

5.1 Assumptions (Directional — Assumed)

- Currency: USD.
- Traffic profile: steady with moderate burst (assumed burst factor ×1.2 for peak-sensitive meters).
- Availability target: 99.9%; DR: RTO 4 hours, RPO 30 minutes.
- Performance: latency-sensitive APIs — low-latency zone placement, premium ingress assumed.
- Pricing basis: public list pricing patterns and meter families (directional, non-contractual).
- Usage volumes assumed:
- Kubernetes workers: 3 × m5.xlarge (4 vCPU / 16 GB), 730 hrs/month, +20% burst headroom.
- Ingress traffic: 18 TB/month egress.
- Messaging: 80 M queue requests/month; 25 M publish+delivery events/month.
- WAF inspected requests: 15 M/month.
- EFS shared storage: 4 TB-month.
- Backup object storage: 8 TB-month + 2 TB/month cross-region replication.
- Secrets and key operations: 25 active secrets, 12 keys, 4 M API operations/month.
- Observability: 50 GB/month log ingest; 40 custom metrics retained 30 days.

5.2 30-Day Total Cost Table (Directional)

Capability	AWS US (baseline, USD)	AWS EU (USD)	AWS AU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Compute/Container	3,980	4,230	4,760	3,760	4,050	4,520	3,860	4,190	4,690	Medium

Capability	AWS US (baseline, USD)	AWS EU (USD)	AWS AU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Networking/Edge	2,420	2,640	2,960	2,280	2,510	2,860	2,360	2,590	2,940	Medium
Messaging	1,090	1,180	1,320	980	1,070	1,230	930	1,030	1,190	Medium
Data/Storage/Backup	3,120	3,390	3,850	2,980	3,240	3,730	3,050	3,320	3,810	Medium
Security/Identity	1,630	1,760	1,980	1,560	1,690	1,920	1,590	1,720	1,950	Medium
Observability	2,760	2,980	3,360	2,640	2,850	3,260	2,700	2,930	3,330	Low
Total (USD)	**15,000**	**16,180**	**18,230**	**14,200**	**15,410**	**17,520**	**14,490**	**15,780**	**17,910**	**Medium**
Delta % vs AWS	**0.0%**	**0.0%**	**0.0%**	**-5.3%**	**-4.8%**	**-3.9%**	**-3.4%**	**-2.5%**	**-1.8%**	**Medium**

5.3 Metered Billing Tier Table (Directional)

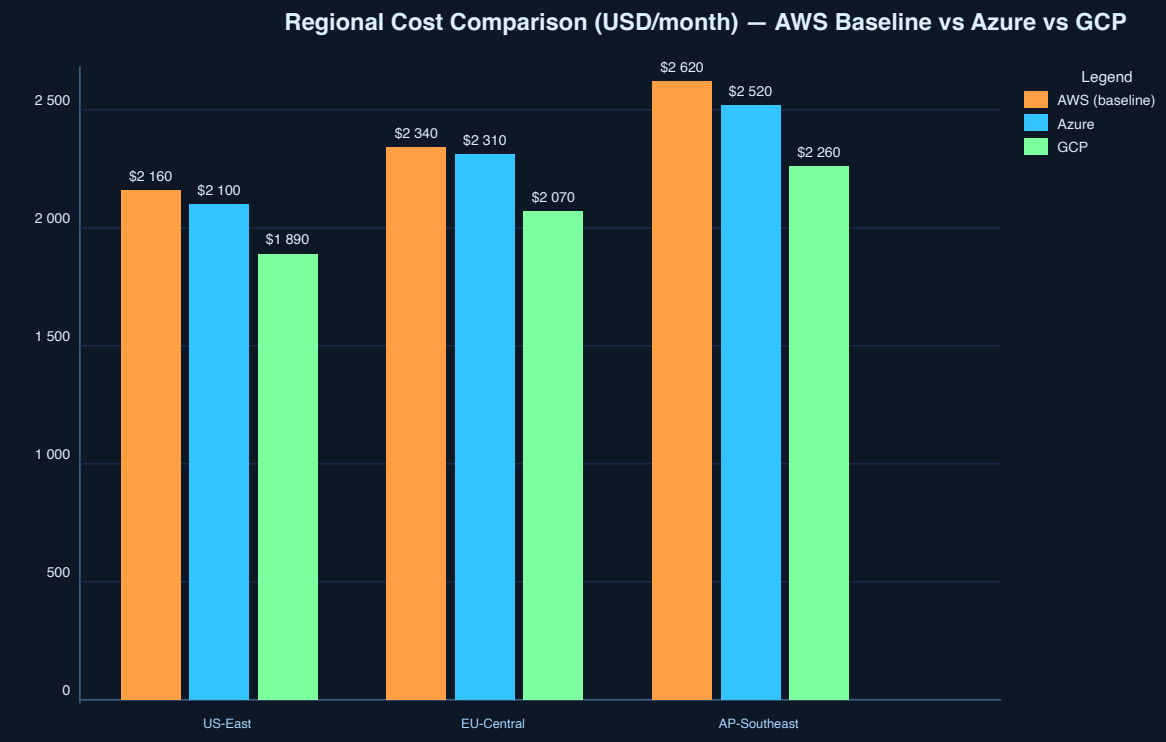
Service	Metering unit	Tier/Band	AWS US (baseline, USD)	AWS EU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
EKS/AKS/GKE worker (m5.xlarge equiv.)	vCPU-hour	First 8,760 vCPU-hrs	1,402	1,521	1,331	1,443	1,594	1,366	1,485	1,628	Medium
Managed K8s control plane	cluster-hour	First 730 hrs	73	79	0 (bundled)	0 (bundled)	0 (bundled)	73	79	86	Medium
Queue ops (SQS/Service Bus/Pub/Sub)	requests	First 1 M	0.40	0.44	0.05	0.06	0.07	0.00	0.00	0.00	Medium
Queue ops (SQS/Service Bus/Pub/Sub)	requests	Over 1 M to 80 M	31.60	34.70	25.20	28.10	32.20	24.00	26.80	30.90	Medium
Topic fanout (SNS/Event Grid/Pub/Sub)	events	First 1 M	0.50	0.55	0.60	0.67	0.77	0.40	0.45	0.52	Low
Topic fanout (SNS/Event Grid/Pub/Sub)	events	Over 1 M to 25 M	12.00	13.20	14.40	16.20	18.10	10.60	11.90	13.80	Low
Shared file storage (EFS/Files/Filestore)	GB-month	First 4,096 GB-month	1,228	1,335	1,150	1,255	1,454	1,190	1,300	1,490	Medium
Backup object storage (S3/Blob/GCS)	GB-month + replication GB	8,192 GB-month + 2,048 GB replication	276	302	251	278	322	259	286	330	Medium
WAF request inspection	requests	First 10 M	60	66	55	61	70	58	64	73	Low
WAF request inspection	requests	Over 10 M to 15 M	25	27	22	25	29	23	26	30	Low

5.4 One-Time Migration Cost Versus Run-Rate Table (Directional)

Cost segment	AWS (baseline, USD)	Azure (USD)	GCP (USD)	Confidence
Platform landing zone + policy baseline uplift	0	28,000	31,000	Medium
IAM/KMS/Secrets model translation and validation	0	18,000	20,000	Medium
Messaging migration (SNS/SQS → target)	0	24,000	26,000	Medium
EKS workload migration, tuning, cutover hardening	0	42,000	46,000	Medium
DR rehearsal and residency validation evidence pack	0	22,000	24,000	Medium
Total one-time migration (USD)	**0**	**134,000**	**147,000**	**Medium**

Cost segment	AWS (baseline, USD)	Azure (USD)	GCP (USD)	Confidence
30-day run-rate (USD, US region)	**15,000**	**14,200**	**14,490**	**Medium**

5.5 Regional Cost Comparison Chart



Azure Δ vs AWS: US −3%, EU −1%, AU −4%
GCP Δ vs AWS: US −13%, EU −11%, AU −14%

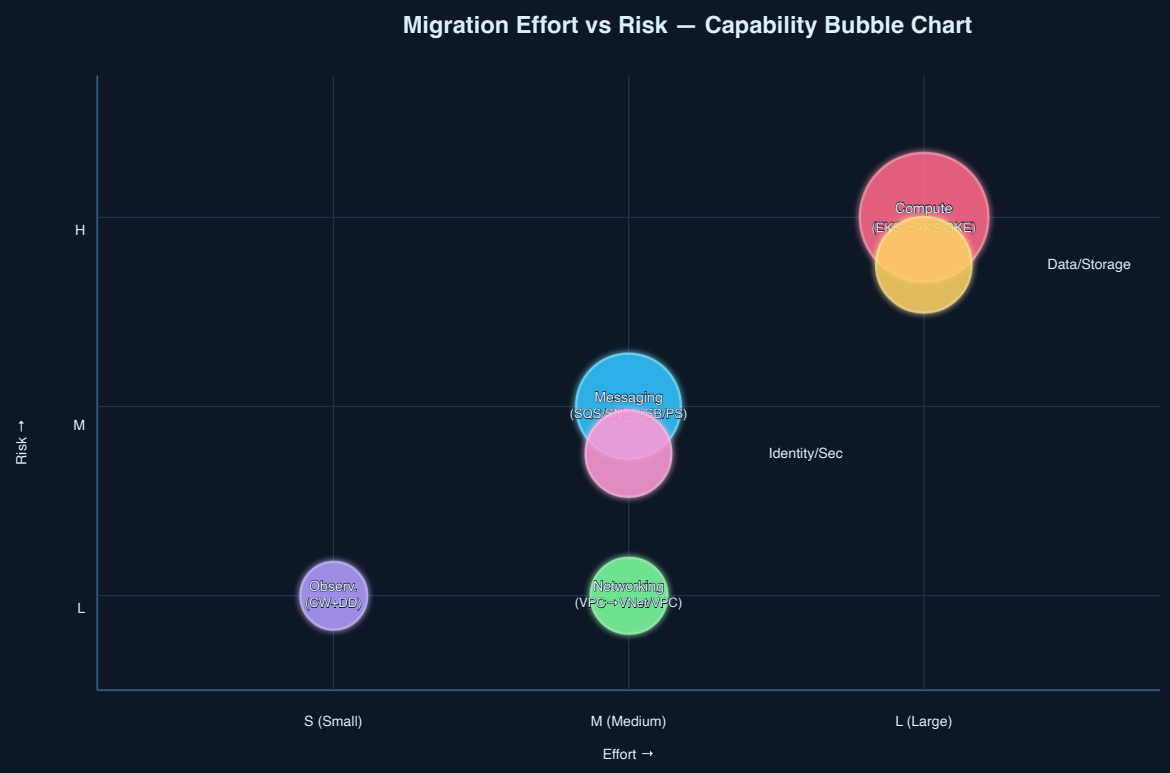
* All values are directional estimates in USD. Confidence: Medium.

6. Migration Challenge Register

Challenge	Impact	Likelihood	Mitigation	Owner role
IRSA → target cloud workload identity translation	High	High	Build identity mapping matrix; implement least-privilege service account roles; stage in non-prod first	Platform Security Lead
SQS/SNS semantics drift in target messaging	High	Medium	Contract tests for ordering, retries, DLQ, and filter policies before production cutover	Integration Architect
Karpenter / autoscaling behaviour parity	Medium	Medium	Replay burst traffic traces in performance environment; tune autoscaler thresholds	SRE Lead
WAF managed rule equivalency gaps	Medium	Medium	Dual-run WAF in monitor mode until false-positive burn-in complete, then enforce	Edge Security Engineer
Backup restore RPO evidence (RTO 4 h, RPO 30 m)	High	Medium	Timed restore drills; document audit evidence for SOC2	DR Manager
Team operating model shift (AWS-native → Azure/GCP-native)	Medium	High	Role-based enablement; paired operations for first two releases	Engineering Manager

7. Migration Effort View

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies
Compute and orchestration (EKS → AKS/GKE)	L	H	Identity, network policy, autoscaling, container registry
Messaging (SNS/SQS → Service Bus/Pub/Sub)	M	H	Contract compatibility, consumer updates, DLQ strategy
Storage and backup (EFS/S3 → Files/Blob or Filestore/GCS)	M	M	Throughput benchmarking, backup restore rehearsal
Security and secrets (KMS/Secrets/IAM)	L	H	RBAC model, key rotation controls, compliance sign-off
Edge and networking (ALB/WAF/Route53/PrivateLink)	M	M	DNS cutover plan, WAF rule parity, private connectivity
Observability (Datadog/CloudWatch)	M	M	Alert parity, SLO burn-rate calibration, runbook updates



8. Decision Scenarios

Cost-first scenario

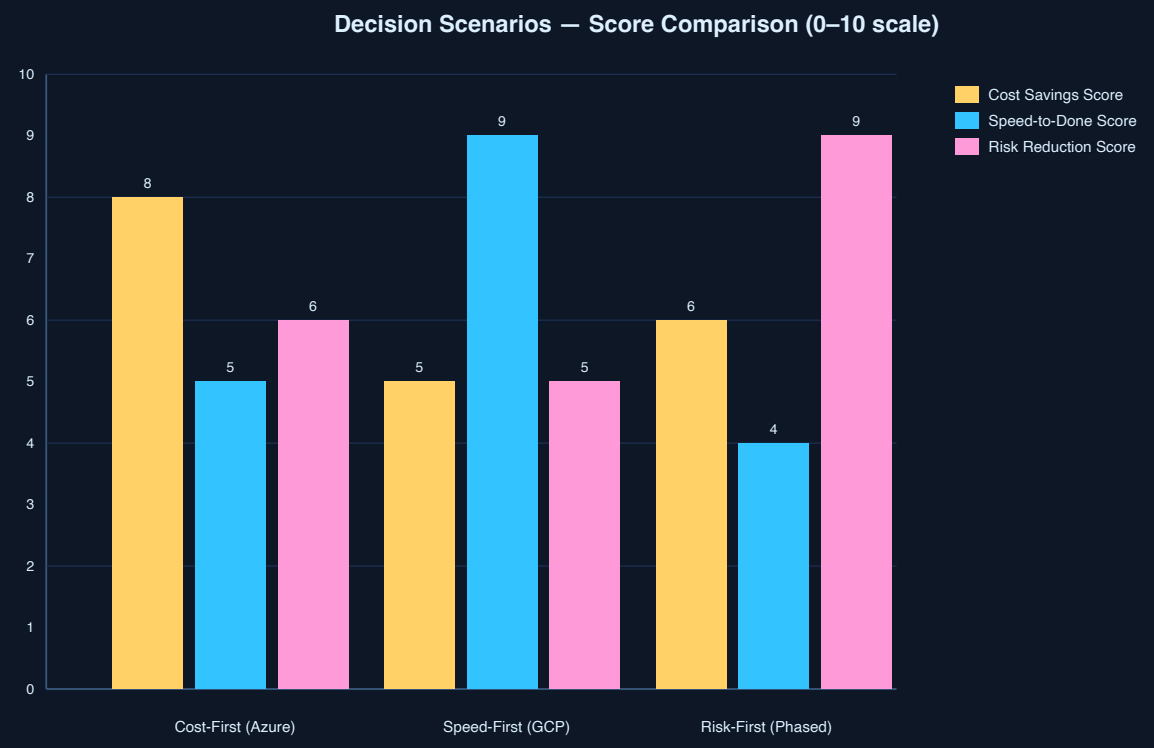
Move messaging and storage first to Azure-native services while keeping compute on AWS temporarily, then migrate AKS in phase 3. This minimises month-1 run-rate quickly but creates temporary cross-cloud integration overhead and complicates identity controls during the transition.

Speed-first scenario

Lift EKS platform directly to AKS with minimal refactor and defer deep optimisation. Fastest delivery to target cloud but carries post-migration tuning debt, higher initial operational noise, and less-proven DR evidence at go-live.

Risk-first scenario

Dual-run non-prod plus canary production slices, complete DR drills and compliance gates before full cutover. Slowest overall path but strongest availability continuity, SOC2 evidence quality, and identity/residency control confidence.



Scores are directional rankings across the three optimisation dimensions (0=worst, 10=best).

9. Recommended Plan (Dynamic Timeline)

Selected timeline: **30/60/90/120 days**.

Rationale: service count and dependency depth are medium-high (EKS core + ingress + WAF + messaging + backup + IAM/KMS/secrets). Data migration complexity is moderate (backup/object and shared file workloads only — no relational engine migration in scoped IaC). Risk profile is high for identity, messaging semantics, and latency-sensitive API behaviour. Four phases are required to achieve SOC2-quality DR evidence and policy-as-code parity before full production cutover.

Phase 1 (Day 0–30): Foundation and parity design

- Objectives: establish target landing zone and reference architecture; agree security and residency design.
- Key activities: identity/key/secrets mapping; networking and ingress design; residency control mapping (Entra/GCP IAM); baseline SLO definitions; non-prod environment provisioning.
- Exit criteria: approved target architecture decisions; security sign-off on identity model; tested non-prod connectivity via PrivateLink/PSC.

Phase 2 (Day 31–60): Non-production migration and contract validation

- Objectives: prove workload function in target cloud with production-like traffic replay.

- Key activities: deploy AKS/GKE equivalent stack; migrate messaging contracts and DLQ patterns; implement WAF rules in monitor mode; validate observability parity (alerts, SLO dashboards); first Velero backup/restore drill.
- Exit criteria: non-prod latency SLOs met; message contract test suite passes; DR backup/restore drill documents RTO ≤ 4 h / RPO ≤ 30 m.

Phase 3 (Day 61–90): Production readiness and progressive cutover

- Objectives: reduce production cutover risk through incremental traffic slices.
- Key activities: canary routing (5 % → 25 % → 50 %); autoscaling/KEDA tuning under production traffic; WAF enforce mode activation; runbook updates; operational handover rehearsal.
- Exit criteria: canary SLOs stable for two full release cycles; incident response drill passes; SOC2 evidence package draft complete.

Phase 4 (Day 91–120): Full cutover and hardening

- Objectives: complete migration and stabilise production operations.
- Key activities: 100 % traffic migration; final DNS/edge cutover; timed DR exercise for RTO/RPO evidence sign-off; cost optimisation pass (right-sizing, reserved capacity); AWS decommission plan.
- Exit criteria: production fully on target cloud; RTO ≤ 4 h and RPO ≤ 30 m evidence signed off; AWS decommission plan approved by stakeholders.

Required architecture decisions before execution

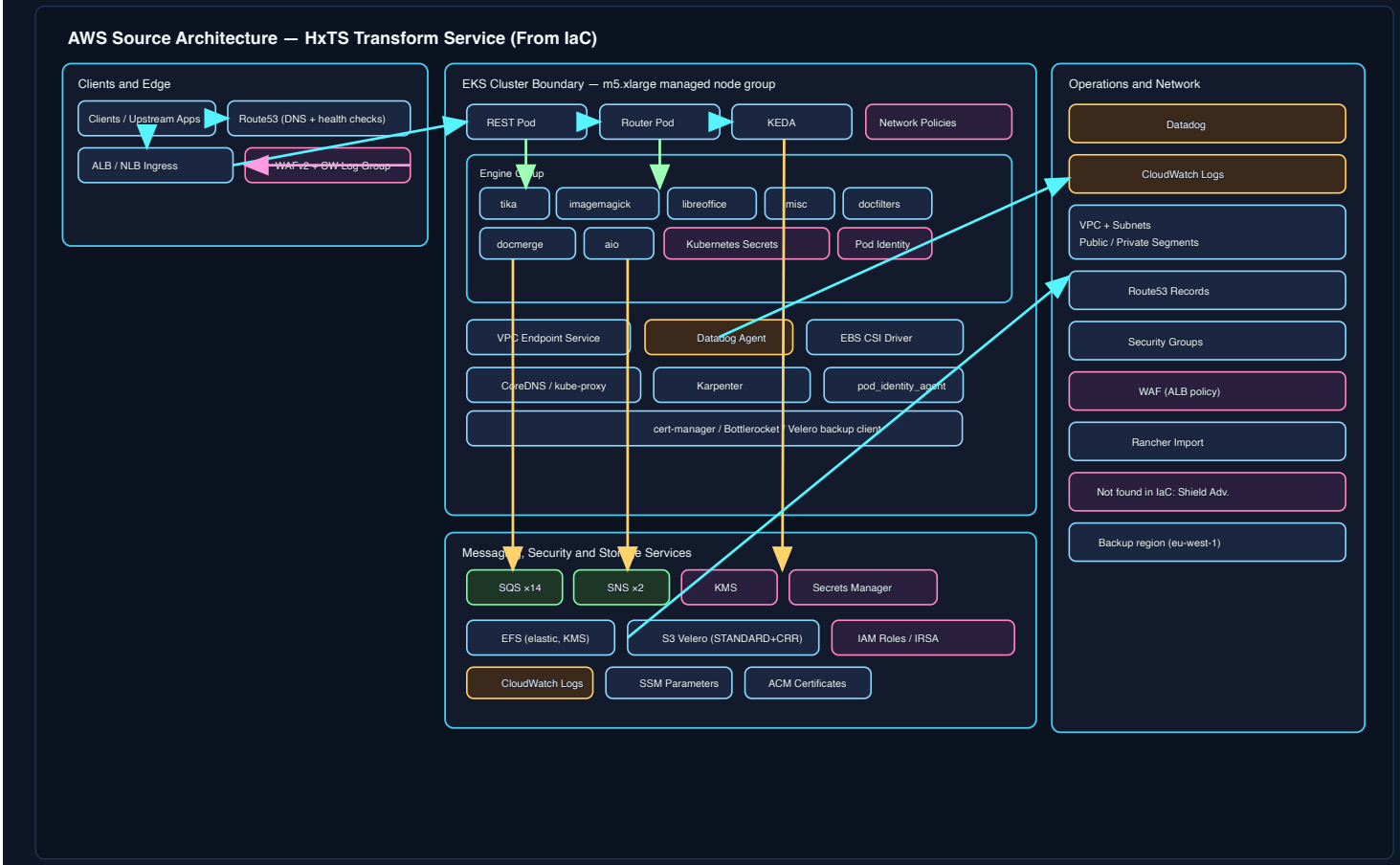
1. Final messaging pattern selection: Service Bus topics vs Event Grid on Azure; Pub/Sub topology on GCP.
2. File storage performance tier for latency-sensitive transforms (Azure Files Premium NFS vs Filestore Enterprise).
3. Workload identity standard and secret rotation approach across all engine deployments.
4. WAF policy set, exception governance model, and logging retention obligations.

10. Open Questions

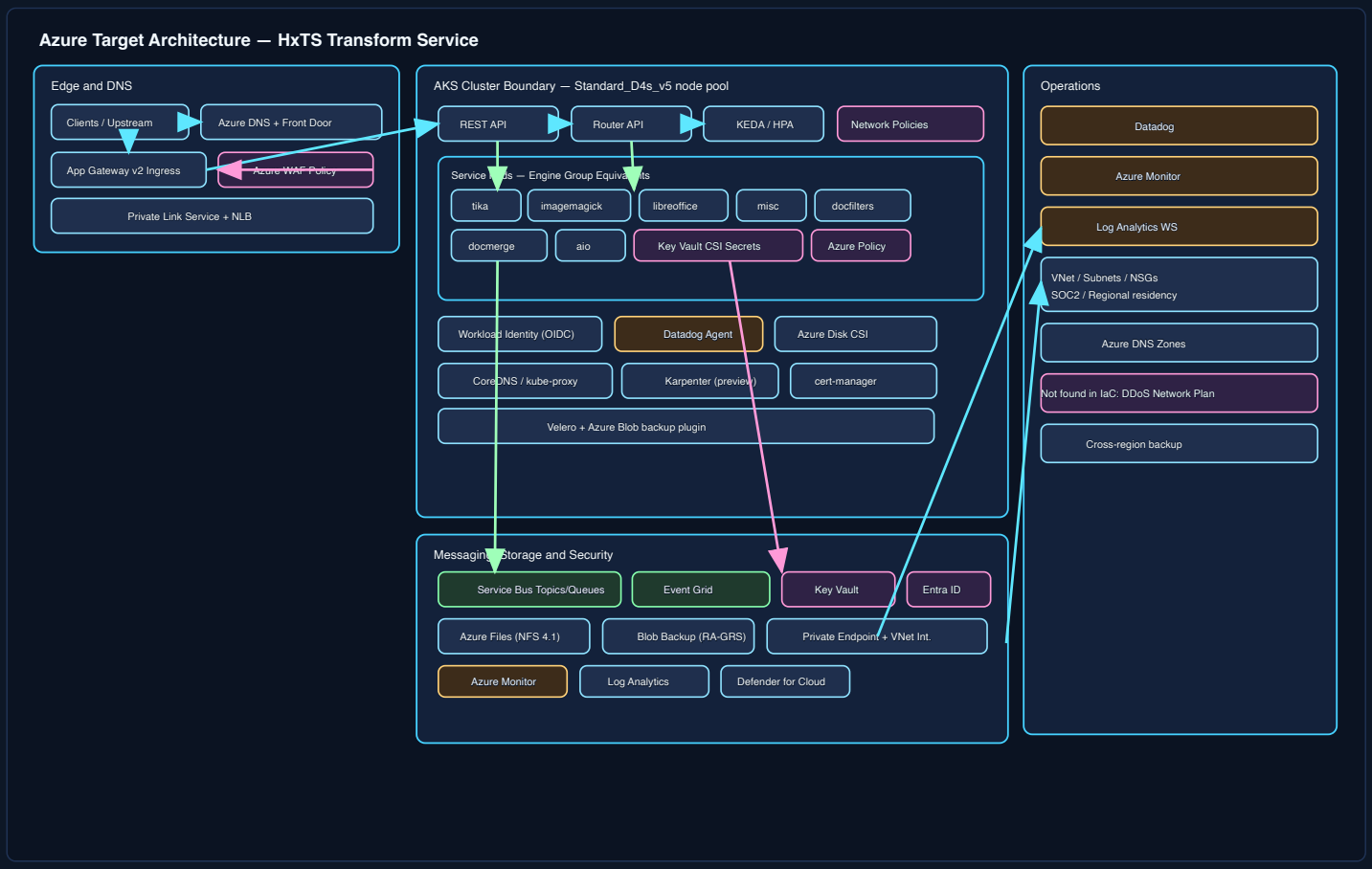
1. What are measured p95/p99 latency SLO values per API route, to calibrate target autoscaling and ingress policy?
2. What are actual monthly queue/event volumes per message type, to replace assumed meter volumes in section 5?
3. Are there legal constraints requiring in-country data residency beyond region-level controls for any tenant?
4. Does DR operate active/passive only, or must selected APIs support active/active?
5. Which observability stack components are contractually required to remain unchanged versus re-platform?

11. Component Diagrams

AWS Source architecture:



Azure Target architecture:



GCP Target architecture:

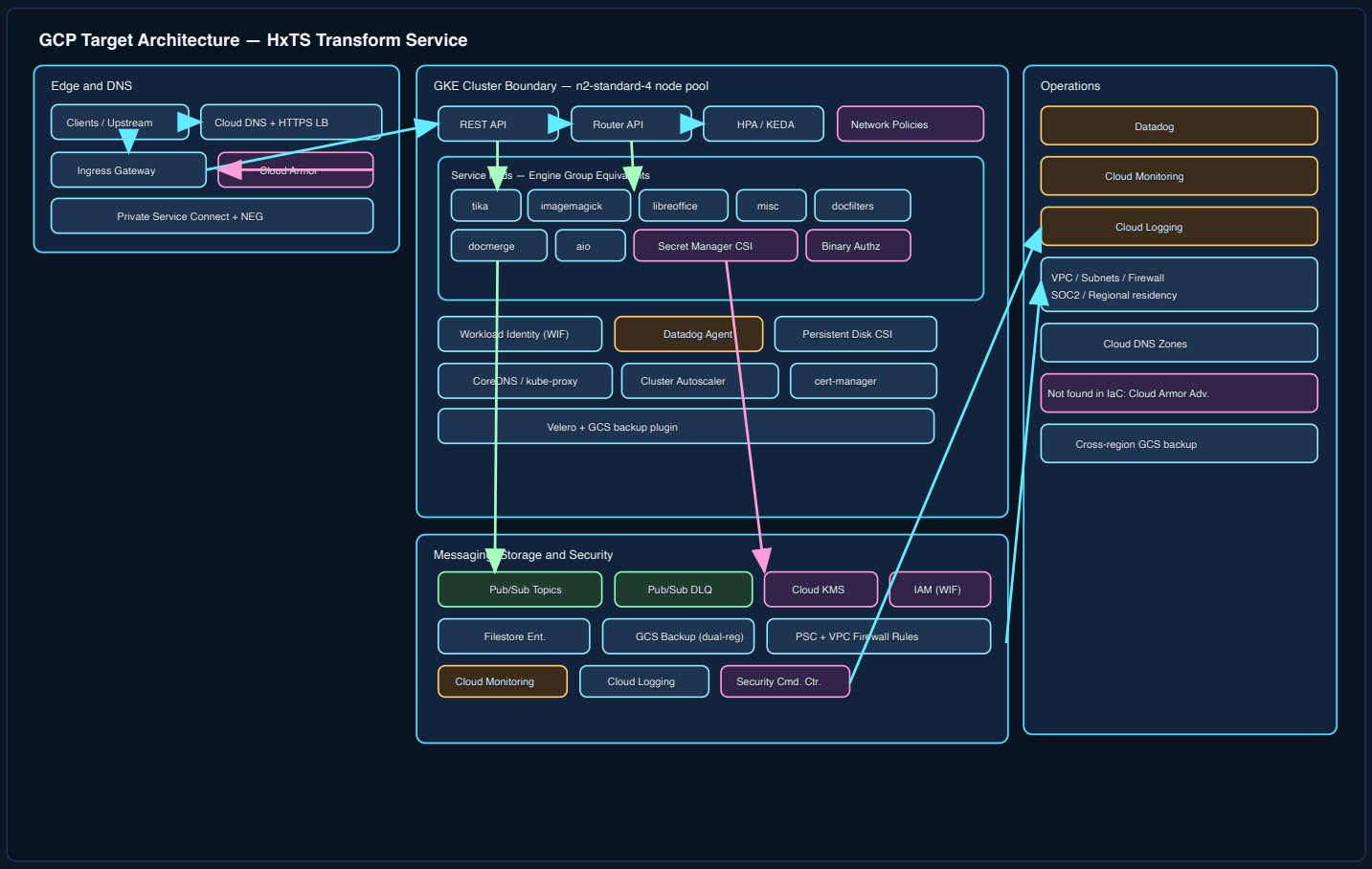


Diagram Legend

- **AWS Source**: clients/upstream → DNS (Route53) → Ingress ALB + WAFv2 → EKS boundary (REST, Router, Engine group: tika/imagemagick/libreoffice/misc/docfilters/docmerge/aio, KEDA) → messaging (SQS/SNS) → security (KMS/Secrets Manager) → observability (Datadog/CloudWatch) → VPC/subnets. Components not found in IaC marked explicitly.
- **Azure Target**: clients → Azure DNS + Front Door → App Gateway + WAF → AKS boundary (equivalent service pods, KEDA/HPA, network policies, Key Vault CSI) → Service Bus/Event Grid → Key Vault + Entra ID → Azure Files/Blob backup → Monitor/Datadog → VNet/subnets.
- **GCP Target**: clients → Cloud DNS + HTTPS LB → Cloud Armor → GKE boundary (equivalent service pods, HPA/KEDA, network policies, Secret Manager CSI) → Pub/Sub → Cloud KMS + IAM → Filestore/GCS backup → Cloud Monitoring/Datadog → VPC/subnets.
- **Supplemental visuals**: cost comparison chart embedded in section 5.5; effort-risk chart embedded in section 7; scenario comparison chart embedded in section 8.